

Vivian Reflection

For this project, I primarily worked on the process-based implementation alongside Anushka, while Grayson and Hannah focused on the thread-based version. Even though we split the implementations, the project still felt highly collaborative because we worked together during class sessions and regularly discussed synchronization strategies, testing approaches, and debugging issues as a group.

Anushka and I used a pair programming style throughout most of the process implementation. We worked on the same laptop during OS class sessions and alternated between writing code and guiding the overall algorithm design and synchronization strategy. This approach helped us catch mistakes early and think more carefully about concurrency problems before implementing them. One person would focus on coding details and syntax while the other checked logic, identified race conditions, potential deadlocks and considered edge cases such as resource contention and starvation.

I think our group worked well together overall. Grayson and Hannah handled the thread implementation effectively, and our discussions comparing the two approaches helped all of us better understand the advantages and disadvantages of threads versus processes. Threads were generally simpler and more efficient because of shared memory, while processes provided stronger isolation at the cost of more complicated communication and synchronization mechanisms.

As a contributor, I mainly focused on the algorithm design and synchronization logic during pair programming sessions rather than coding, partially because my gaming laptop battery dies faster than Anushka's. I also helped analyze synchronization behavior and verify that the system avoided getting stuck under different parameter combinations, especially shorter times for cats to be satiated.

One thing I improved on during this project was thinking more systematically about concurrency issues before coding. I am usually more comfortable diving directly into implementation and refining the system design while coding, testing, and debugging. However, this project pushed me to slow down and think more carefully about synchronization requirements, critical sections, and potential deadlock situations before writing code.

Overall, this project gave me a much stronger understanding of synchronization, concurrency, and the practical differences between thread-based and process-based designs. Working collaboratively in a pair programming environment also made debugging and problem solving more effective because we were constantly reviewing each other's reasoning and implementation decisions.

